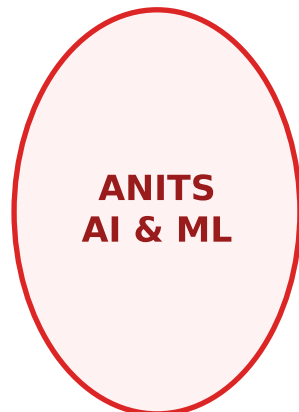


Retail Multi-Segment Profiler & High-Value Customer Classifier

Report submitted as part of the curriculum program requirement for the degree of
**BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE ENGINEERING (AI & ML)**

Submitted by

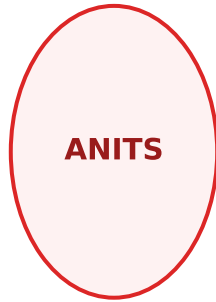
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2023-2027



BONAFIDE CERTIFICATE

This is to certify that this Internship Report "Retail Multi-Segment Profiler & High-Value Customer Classifier" from

This work embodies authentic research and software implementation in Unsupervised Persona Clustering, Super

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1. Introduction

1.1 Background of Internship

This report highlights my learning experience and technical outcomes from a four-week virtual internship, undertaken at [Company Name].

The program was designed around project-based experiential learning and industrial mentorship. The core objective was to provide hands-on exposure to real-world challenges in retail analytics.

In modern omnichannel commerce, understanding customer purchasing behavior and identifying high-value shoppers are critical for driving growth and optimizing marketing spend.

1.2 Objectives of the Internship

The primary objective of this internship was to gain hands-on expertise in developing, benchmarking, and deploying machine learning models for customer segmentation and high-value classification.

- To understand the business mechanics of customer segmentation and predictive lifetime value modeling.
- To perform comprehensive data preprocessing, outlier detection, and leakage-free feature engineering.

- To apply K-Means Clustering on customer demographic and spending dimensions, evaluating cluster optimality.
- To profile discovered clusters into actionable business personas (e.g. Affluent VIP Spenders, Conservative Savers).
- To train and benchmark Supervised Classifiers (Logistic Regression, Decision Trees, Random Forests, and XGB).
- To prevent data leakage by isolating behavioral predictors from target label formulations.
- To develop and deploy an interactive Streamlit Web Application featuring dynamic Plotly 2D/3D charts, a real-time dashboard, and a feedback loop.

1.3 Importance of AI Technologies in Retail Analytics

Artificial Intelligence is revolutionizing retail customer relationship management (CRM). As datasets grow exponentially, AI enables retailers to analyze vast amounts of data to identify patterns, predict customer behavior, and personalize marketing campaigns.

1.4 Scope of the Project

The scope of this project encompasses building a complete, reproducible machine learning pipeline starting from raw data collection to model deployment and monitoring.

2. Organization Profile

2.1 AICTE Virtual Internship Overview

The All India Council for Technical Education (AICTE), as the statutory apex body for technical education in India, pro

2.2 Role of Edunet Foundation

The Edunet Foundation operates as the primary talent development and project execution partner. As a distinguish

2.3 Shell as the Industry Sponsor

Shell, a global leader in energy, digital transformation, and industrial analytics, served as the industry sponsor for t

Summary of Internship Program Details

The following table outlines the key operational, organizational, and technical dimensions:

Program Dimension	Institutional Details & Specifications
Institutional Partners	AICTE (Regulatory Body), Edunet Foundation (Execution), Shell (Sponsor)
Program Title	Skills4Future Virtual Internship: AI & Data Analytics for Customer Intelligence
Program Duration	4 Weeks (Intensive Project-Based Learning & Mentorship)
Delivery Mode	Fully Online / Remote (GitHub, Colab, Python SDK, Streamlit Cloud)
Eligibility Criteria	B.Tech Computer Science Engineering / AI & ML Specialized Undergraduates
Core Technologies	Python 3.14, Pandas, NumPy, Scikit-Learn, XGBoost, SHAP, Plotly, Streamlit
Key Deliverables	Data Pipeline, K-Means Clustering, XGBoost Classifier, Streamlit Dashboard
Stipend & Fees	Zero-Cost Program with Merit-Based Industrial Certification
Academic Credit	Formally recognized academic internship credit fulfilling degree requirements

3. Project Overview

3.1 Title of the Project

Retail Multi-Segment Profiler & High-Value Customer Classifier
(Machine Learning & Customer Intelligence System for Retail Decisioning)

3.2 Problem Statement

Modern retailers face intense market competition, rising customer acquisition costs (CAC), and declining margins from

1. Group thousands of diverse shoppers into distinct, interpretable buying personas without manual rules.
2. Accurately predict which customers will emerge as High-Value shoppers based on behavioral spending signals.
3. Generate tailored marketing strategies that maximize Customer Lifetime Value (CLV) without revenue leakage.

3.3 Objectives of the Project

- Unsupervised Segmentation: Deploy K-Means clustering to discover 5 core retail customer personas.
- Cluster Evaluation: Determine optimal K using Within-Cluster Sum of Squares (Inertia) & Silhouette Scores.
- Supervised Classification: Train an XGBoost classifier to identify high-value customer propensity.
- Target Leakage Elimination: Separate predictive features from ground-truth value labels.
- Interactive Dashboard: Deploy a Streamlit web application for real-time customer profiling and strategy execution.

3.4 Relevance to Retail Multi-Segment Intelligence

This project provides direct business ROI by transforming raw transaction logs into proactive CRM action plans. By c

4. Literature Review / Theoretical Background

4.1 Introduction to Customer Segmentation & Behavioral Dimensions

Customer segmentation is the analytical process of dividing a broad consumer market into distinct subsets of consumers.

The classic Mall Customer dataset focuses on three foundational attributes:

- Age: Reflects generational life-stage, disposable income priorities, and digital channel affinity.
- Annual Income (\$k): Quantifies the absolute financial capacity and purchasing power of the shopper.
- Spending Score (1-100): A synthetic behavioral index computed from transaction frequency, basket size, and brand loyalty.

While demographic data provides baseline context, modern retail intelligence incorporates transactional variables such as purchase frequency and basket size.

4.2 Role of Unsupervised Learning & K-Means Clustering

Unsupervised machine learning algorithms uncover hidden geometric structures in unlabelled multidimensional features.

Given a dataset $\{x_1, x_2, \dots, x_n\} \in \mathbb{R}^d$, K-Means partitions the observations into K clusters $S = \{S_1, S_2, \dots, S_K\}$ by minimizing the within-cluster sum of squares.